

Conditioning or Orientation? Shrinkage Targets for Characteristic-Managed Portfolios

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Abstract

Covariance shrinkage has become the standard response to the instability of high dimensional covariance estimation. But there isn't substantial work on shrinkage targets developed for the specific structure of the assets being studied rather than as general purpose regularizers. In this paper I propose the second moment of the characteristics themselves as a prior on the estimation of the covariance of characteristic-managed portfolios, aided by the symmetry in their dimensions. The motivation for the target is both its greater estimation stability and a mechanical connection between characteristic overlap and managed return covariance. Out of sample, my proposed methodology ranks first among the 23 methods and benchmarks tested. Performing a matched spectrum eigenvector rotation experiment to decompose the channels through which this performance gain operates shows that most of it is associated with regularization rather than the characteristic directions. However, a general scaled identity prior carrying no characteristic information at all reaches nearly the same Sharpe performance which weakens the justification for adding detailed economic structure to the prior. The measurement of the mechanism itself is what turns that negative result into an informative methodological note.

1 Introduction

Characteristic-managed portfolios turn a large cross section of security-level information into a span of tradable payoffs in the asset universe. Given a set of characteristic-managed portfolio returns to combine efficiently, the natural portfolio-choice candidate is the maximum Sharpe Ratio portfolio combination within that span. This efficient direction in the portfolio weight space is defined as: $b^* = \Sigma_f^{-1} \mu$ and is the object I study. To get to b^* , we need a good estimate of the covariance matrix and subsequently its inverse Σ_f^{-1} .

The obvious statistical procedure would be to estimate μ and Σ_f and then invert the estimated Σ_f . However, it has been well studied that inverting an already estimated matrix reweighs towards the smallest eigenvalues, where the estimation error is the largest [El Karoui, 2010].

The response from the literature is to shrink the estimate of the covariance matrix towards a structured target that behaves well [Kozak et al., 2020, Ledoit and Wolf, 2004, 2020]. This motivates my central scientific question: **what target should we shrink towards and why?**

The structured covariance shrinkage target I explore in this paper is the second moment of the characteristics themselves, called Σ_x here. There are a few hypothetical reasons to consider it as a shrinkage target for covariance estimation.

A naive one is simply that Σ_x is estimated from far more data than Σ_f . Σ_x has a cross section of information across all assets available every month while Σ_f has a single observation per month per characteristic to estimate from. The difference is in orders of magnitude and may result in vastly different precision of estimation of the two objects. Another hypothetical reason that may be considered is the conditioning of the two matrices. I find empirically that Σ_f is far more ill-conditioned than Σ_x . Conditioning and precision by themselves may not be a good reason to use Σ_x since a well-estimated and stable random matrix with no economic relevance may not be a good shrinkage target. Thus, another motivating reason is mechanical: Σ_f is estimated from portfolios that are themselves built from characteristics since managed returns f_t are built from characteristic levels. So Σ_x is not just a precise but unrelated matrix; Σ_f may inherit some of the structure of Σ_x since portfolios built from correlated characteristic weights hold overlapping positions in assets that co-move.

These must be tested with well-defined experiments however. Prior work motivates that characteristics can be related to loadings, like in the IPCA work done by KPS [Kelly et al., 2019].

Among the 23 methods and benchmarks tested, the portfolio formed from covariance estimation shrinkage towards Σ_x ranks first in Sharpe, significantly ahead of some and statistically tied with others. It beats popular methods like RIE [Bun et al., 2016] and the tuned POET [Fan et al., 2013]. 12 of the 22 comparisons survive Bonferroni corrections at $\alpha = 0.05/22 = 0.00227$. It is important to specify that the result holds only under the scope the paper defines, and not for general covariance estimation.

Using Σ_x as a shrinkage target tries to address two issues with directly estimating Σ_f at the same time. Σ_x has a stable spectrum and is substantially better conditioned than Σ_f , and its characteristic specific eigenvectors give us characteristic directions.

This creates an additional confound to disentangle as any gain over a plug-in target may come from either source. Thus the research questions I define are: **Do the directions of particular characteristics carry information?** Or is any well behaved, stable target sufficient for the purpose of regularization? The two channels are not mutually exclusive. So **how do we separate and quantify the effect of each?**

The practical step to separate the two effects is to hold one constant and vary the other. I do this by holding the eigenvalues fixed, while rotating the eigenvector directions randomly. This creates a separation of the effect of the conditioning of the matrix and the directions it carries in return space. This is made more precise in the design section.

The results of this design are 2.426 (plug-in) $\rightarrow$ 2.865 (rotation mean) $\rightarrow$ 3.056 (Σ_x). I call the first gap the conditioning component and the second the orientation component, and define both in the design section.

The two targets behind the second gap share a spectrum and differ in their eigenvectors, though

the analytical rule gives them different realised intensities.

While stabilization of the matrix being inverted accounts for about 70% of the sum of the two operational gaps, the increment from characteristic directions is statistically detectable and not negligible.

On testing 3 different targets for absolute Sharpe performance instead of incremental gains, I find that Σ_x , the scaled identity $I \times \text{tr}(\Sigma_x)/p$, and the model implied mechanical benchmark B_{mech} , defined in the results section, all land within 0.0155 of each other.

This work is related to two literatures in the empirical asset pricing space:

The shrinkage/regularization literature asks how to estimate covariance better [Ledoit and Wolf, 2004, 2020, Bun et al., 2016, Kozak et al., 2020]. The characteristics/factor literature asks what characteristics organize return covariance [Fama and French, 1993, Kelly et al., 2019].

I ask a narrower question at the intersection of the two: **how much of the performance of a covariance target comes from its characteristic-specific orientation, rather than generic regularization?**

The paper makes two contributions.

- An empirical comparative analysis of various covariance shrinkage methodologies from the literature for application to managed portfolios. This is where the proposed scheme using Σ_x ranks first, with 12/22 surviving Bonferroni corrections.
- A design that operationally decomposes the two channels by holding the spectrum of Σ_x constant and varying its eigenvectors by randomized orthogonal draws.

This work relates to two recent results:

- The scaled identity as a benchmark and the decomposition of Sharpe gains into conditioning and orientation relates directly to the Rotational Invariant Estimator work [Bun et al., 2016] where they keep the eigenvectors fixed and optimally re-estimate eigenvalues. I show that by using Σ_x , we can add the orientation effect as a prior belief on the eigenvector structure, and that this is not negligible in the data. In their conclusion they name a factor-model prior on the eigenvectors as an open problem. For this application, Σ_x beats the whitened RIE estimator with a p-value of 7.7×10^{-4} .
- Goldberg, Gurdogan and Kercheval’s recent work [Goldberg et al., 2025] shows analytically that the performance improvement from their eigenvector shrinkage depends on two quantities: the angle between the population eigenvector and the shrinkage target, and the relative gap between the leading sample eigenvalue and the rest. My rotation experiment is somewhat related but asks a different question: how much performance survives once the characteristic orientation is removed and the spectrum is held fixed?

2 Setup

Notation. I now define precisely the objects involved in the analysis. First, data count notation:

- $p = \# \text{ characteristics} = \# \text{ managed portfolios (one per characteristic)} = 132$ of 153 available in JKP US equities dataset [Jensen et al., 2023].¹ All 132 characteristics selected are linearly independent/full rank.
- $T = \# \text{ months in the sample} = 755$.²
- $N_t = \# \text{ assets/stocks in the cross section of month-}t \text{ with } E[N_t] = 3422$.

Now define characteristic-managed portfolio returns as:

$$f_t = \frac{X_t' r_{t+1}}{N_t} \in \mathbb{R}^p$$

where:

- $r_{t+1} \in \mathbb{R}^{N_t} = \text{next month's return vector for this month's selected cross section of stocks (no forward looking bias)}$.
- $X_t \in \mathbb{R}^{N_t \times p} = \text{month-}t \text{ characteristics matrix, cross sectionally z-scored}$.³

Now stack managed portfolio returns f_t' as rows in:

$$F \in \mathbb{R}^{T \times p}$$

with $\mu = E[f_t] \in \mathbb{R}^p$, $\Sigma_f = \text{Var}(f_t) \in \mathbb{R}^{p \times p}$, and $\gamma = p/T = 0.175$ which is the concentration ratio for estimating Σ_f over the full sample [Marchenko and Pastur, 1967], a mild case of γ [Chernov, 2026]. Estimation uses expanding windows, so the ratio the estimator actually faces runs from $132/180 = 0.73$ at the first refit down to 0.175 at the last.

Finally, define characteristics' second moment as:

$$\Sigma_x = E[X_t' X_t / N_t] \in \mathbb{R}^{p \times p}$$

Two estimands. $b^* = \Sigma_f^{-1} \mu$ is a portfolio estimand that asks when given a set of managed portfolio returns, how should we optimally combine them. $\beta^* = \Sigma_x^{-1} \mu$ is a prediction estimand that asks when given a stock's characteristics, what return should we predict.

¹There are some discrepancies in related published work using this dataset in the rules of thumb applied to characteristic selection, imputation of missing values, etc. KKM AIPM (2025) and FKMZ RPM (2026) state a 30% missing threshold and report 132 while attributing to Didisheim et al (DKKM). On this dataset that threshold gives 121. Retaining 132 would require cutting at 35.5%, and 22 of the 153 characteristics sit in [30%, 40%) missing. I therefore use DKKM's rank rule, whose count is stable across data vintages [Kelly et al., 2025, Fan et al., 2026, Didisheim et al., 2024].

²Jan 1963 – Nov 2025; Dec 2025 excluded as no t+1 data available for that month.

³First following AIPM's rank standardization [Kelly et al., 2025], then re-standardized to z-scores each month before moments are taken. This is done so that $\text{diag}(\Sigma_x)$ for characteristic-j is the N-weighted fraction of months in which j has cross sectional variation and Σ_x eigenvalues largely represent correlations among characteristics and not data coverage differences.

Both of these estimands have the same numerator but different metrics and they solve for different problems. In the Hansen-Richard structure, a change in inner product changes which projection of the SDF is recovered [Hansen and Richard, 1987]. They only coincide if Σ_f and Σ_x have the same effect on μ . It is thus important to separate the two second moments since using a single Σ notation collapses the two into a single object.

Now I ask more precisely whether the two metrics have the same linear operator effect on μ in our sample. In the data, running a regression of a constant on the matrix F of managed returns without an intercept recovers the tangency weight directions up to scale as per Britten-Jones [Britten-Jones, 1999]. I then impose proportionality to β^* as $p-1$ linear restrictions i.e. every direction except β^* itself so that the scale is a free variable. An F-statistic of 3.861 on $(p-1 = 131, T-p = 623)$ degrees of freedom, with a p-value of 6.5×10^{-30} , rejects the claim that the two metrics are proportional in the data.⁴

Estimation asymmetry. I mentioned some hypothetical reasons to consider Σ_x as a shrinkage target in the introduction, with the most naive one being a larger sample size amounting to greater precision: $\Sigma_f \in \mathbb{R}^{132 \times 132}$ is built from 755 monthly observations with $\gamma = 0.175$ while Σ_x is built from 2,583,715 stock-months. But the count alone overstates the precision gain and as pointed out, precision for small eigenvalues matters considerably more than that of the top of the spectrum. Under a 400-draw month bootstrap, Σ_f 's eigenvalues have a relative standard error 19.5 times Σ_x 's in the top decile of the spectrum, and 3.2 times in the bottom. Relative estimation error of Σ_f^{-1} and Σ_x^{-1} under a monthly bootstrap with replacement compared to a full sample estimate shows a large gap. Σ_x^{-1} moves 3% on resampling months while Σ_f^{-1} moves 47%, a stability issue that makes Σ_f worth shrinking, and also Σ_x worth shrinking towards. Σ_f 's condition number is 81,738 against Σ_x 's 1,714, so inverting Σ_f amplifies whatever error it carries.

3 Mechanical Identity

In the introduction I suggested that Σ_f may inherit some of the structure of Σ_x which I now make more precise.

Prior work on loadings. Consider the factor model:

$$r_{t+1} = B_t g_{t+1} + \epsilon_{t+1}$$

where:

- K is the number of latent factors.
- $B_t \in \mathbb{R}^{N_t \times K}$ are the month-t loadings.

⁴Exactness holds under multivariate normality of managed returns which isn't supported by the data.

- $g_{t+1} \in \mathbb{R}^K$ are the next month factor returns with $E[g_{t+1}|X_t] = \bar{g}$ and $\text{Cov}(g_{t+1}|X_t) = \Sigma_g$.

There is prior precedent for factor loadings being related to observable characteristics, recently and notably the IPCA work done by KPS [Kelly et al., 2019]. In particular, KPS posit a characteristic driven component, plus a residual loading component. I set that residual to 0 giving:

$$B_t = X_t \Gamma$$

where $\Gamma \in \mathbb{R}^{p \times K}$ is the characteristic-to-loading map. In other words, I assume that characteristics fully explain loadings. This is much stronger than the assumptions made by previous work but serves as a benchmark since the mechanical term is cleanest in this form. A caveat is that anything the residual would have carried is excluded by assumption.

Assumptions. Assumptions are stated for general weights W_t . I derive the identity under general weights and specialize to equal weighting at the end.

- A1: $B_t = X_t \Gamma$ as above.
- A2: $E[\epsilon_{t+1}|X_t] = 0$.
- A3: $\text{Var}(\epsilon_{t+1}|X_t) = \sigma^2 I$.
- A4: $\text{Cov}(\epsilon_{t+1}, g_{t+1}|X_t) = 0$.
- D1: $f_t = X_t' W_t r_{t+1} / N_t$ with weights $W_t = \text{diag}(w_{t,1}, \dots, w_{t,N_t})$ fixed given X_t , so they cannot depend on r_{t+1} .
- A5: in the factor second moment, $A_t = X_t' W_t X_t / N_t$ is replaced by its mean $A = E[A_t]$.
- A6: $C_t = X_t' W_t^2 X_t / N_t$ stays random, with $\text{Cov}(1/N_t, C_t) = 0$ and $C = E[C_t]$.

A2 makes the residual term in μ go to 0, A3 gives the κ term, and A4 makes the cross term in the variance go to 0. D1 generalises the managed return definition from the setup, which is the case where $W_t = I$. I require W_t to be fixed given X_t because every step below conditions on X_t and treats the weights as known when it does, which is what A2 and the conditional moments of g_{t+1} need. A6 says the size of the cross section is uncorrelated with its characteristic overlap. I concede that A5 and A6 are strong assumptions but can measure the error each one introduces. A_t does vary a lot across months as is expected with real world constraints of markets. The mean of $\|A_t - \Sigma_x\|_F / \|\Sigma_x\|_F$ is 0.45, so A5 is not close to exact. The error is smaller in the places where the identity actually uses these assumptions. Replacing A_t by its mean costs 12% in the factor term at $M = \Sigma_x$, and the A6 factorisation leaves $E[C_t/N_t]$ differing from $E[1/N_t] \Sigma_x$ by 23% in magnitude and 13 degrees in orientation.

Without A5 the factor term is $E[A_t M A_t]$ rather than $A M A$ with the time variation in $A_t \Gamma \bar{g}$ contributing as well. Holding A_t at A removes both of these and isolates the mechanical channel.

Substitution. Now substituting the loading model into the managed returns definition i.e. $r_{t+1} = X_t \Gamma g_{t+1} + \epsilon_{t+1}$ in $f_t = \frac{X_t' W_t r_{t+1}}{N_t}$:

$$f_t = \underbrace{\frac{X_t' W_t X_t}{N_t}}_{A_t} \Gamma g_{t+1} + \underbrace{\frac{X_t' W_t \epsilon_{t+1}}{N_t}}_{u_t}$$

where A_t is the characteristic overlap matrix and u_t is the residual component of the managed return.

Mean.

$$\mathbb{E}[u_t] = \mathbb{E} \left[\frac{X_t' W_t \mathbb{E}[\epsilon_{t+1} | X_t]}{N_t} \right] = 0$$

Using $\mathbb{E}[\epsilon_{t+1} | X_t] = 0$ from A2 and the weights being fixed given X_t from D1.

$$\mu = \mathbb{E}[f_t] = \mathbb{E}[A_t \Gamma g_{t+1}] = \mathbb{E}[A_t] \Gamma \mathbb{E}[g_{t+1}] = A \Gamma \bar{g}$$

Using the law of iterated expectations, since D1 makes A_t a function of X_t by itself and $\mathbb{E}[g_{t+1} | X_t] = \bar{g}$ does not depend on X_t .

Variance. Now for the variance:

$$\Sigma_f = \text{Var}(A_t \Gamma g_{t+1}) + \text{Var}(u_t) + 2 \text{Cov}(A_t \Gamma g_{t+1}, u_t)$$

$$\text{Cov}(A_t \Gamma g_{t+1}, u_t) = \mathbb{E} \left[A_t \Gamma \text{Cov}(g_{t+1}, \epsilon_{t+1} | X_t) \frac{W_t X_t}{N_t} \right] = 0$$

from A4, without which the cross term does not vanish and the identity does not reduce.

Using A_t replaced by its mean A from A5 and $M = \Gamma \Sigma_g \Gamma' \in \mathbb{R}^{p \times p}$, symmetric, the factor part reduces to:

$$\text{Var}(A_t \Gamma g_{t+1}) = A \Gamma \text{Var}(g_{t+1}) \Gamma' A = A \Gamma \Sigma_g \Gamma' A = A M A$$

The residual part reduces to:

$$\text{Var}(u_t) = \sigma^2 \mathbb{E} \left[\frac{X_t' W_t^2 X_t}{N_t^2} \right] = \sigma^2 \mathbb{E} \left[\frac{1}{N_t} C_t \right] = \kappa C, \quad \kappa = \sigma^2 \mathbb{E}[1/N_t]$$

From A3 for the first equality, and A6 for the last one.

So under general weights the identity is:

$$\Sigma_f = A M A + \kappa C$$

The identity under equal weighting. I set $W_t = I$ implying an equal weighting construction. Equal weighting is what makes the identity a statement about Σ_x . Under equal weighting both

overlap matrices become the same object, the Σ_x defined in the setup.

$$W_t = I \Rightarrow A_t = C_t = \frac{X_t' X_t}{N_t}, \quad A = C = E \left[\frac{X_t' X_t}{N_t} \right] = \Sigma_x$$

$$\boxed{\Sigma_f = \Sigma_x M \Sigma_x + \kappa \Sigma_x}$$

So under these assumptions Σ_f is a quadratic in Σ_x plus a multiple of Σ_x . This is the mechanical reason given in the introduction. The identity only tells us that Σ_f is in the span of $\Sigma_x M \Sigma_x$ and Σ_x , which is weaker than saying that Σ_x is the right target to shrink towards. I shrink towards Σ_x because it needs no estimate of M , and I test the $\kappa \Sigma_x$ term separately as B_{mech} .

Under general weights the mechanical term involves A and C rather than Σ_x itself. All comparisons here are made under a common weighting.⁵

4 Design

The rotation construction. We can hold the whole spectrum, and hence the condition number, constant while varying the eigenvectors by using randomized orthogonal draws. This gives us a target as well behaved as Σ_x but without any characteristic specific identity that defines its eigenvector directions. In the decomposition $\Sigma_x = Q \Lambda Q'$, replace Q with random orthogonal rotations while keeping Λ the same:

$$\Sigma_x = Q \Lambda Q' \longrightarrow T_j = Q_j \Lambda Q_j', \quad Q_j \text{ random orthogonal, } j = 1, \dots, 2000$$

Estimation protocol. Every estimator here is run on one expanding window with one refit schedule:

- the first 180 months are used to estimate and are never scored. Refits happen every 12 months after that, which leaves 575 out-of-sample months, and every Sharpe ratio I report is computed on those months except where I note otherwise below.
- the plug-in estimator is the sample covariance with no target.
- every estimator is inverted under the same 10^{-4} ridge, and every blended estimator uses the same b^2 on a trace-matched target, so they differ in the target they shrink towards and in the intensity that target implies, which I return to below.
- p-values against Σ_x come from a HAC delta-method test for the difference in Sharpe ratios with a 12 month lag following Ledoit and Wolf on paired out-of-sample months [Ledoit and Wolf, 2008].

⁵NOT Assumed: N_t constant; it varies over months and the identity needs it to. $\kappa = \sigma^2 E[1/N_t]$ is an expectation over exactly that variation.

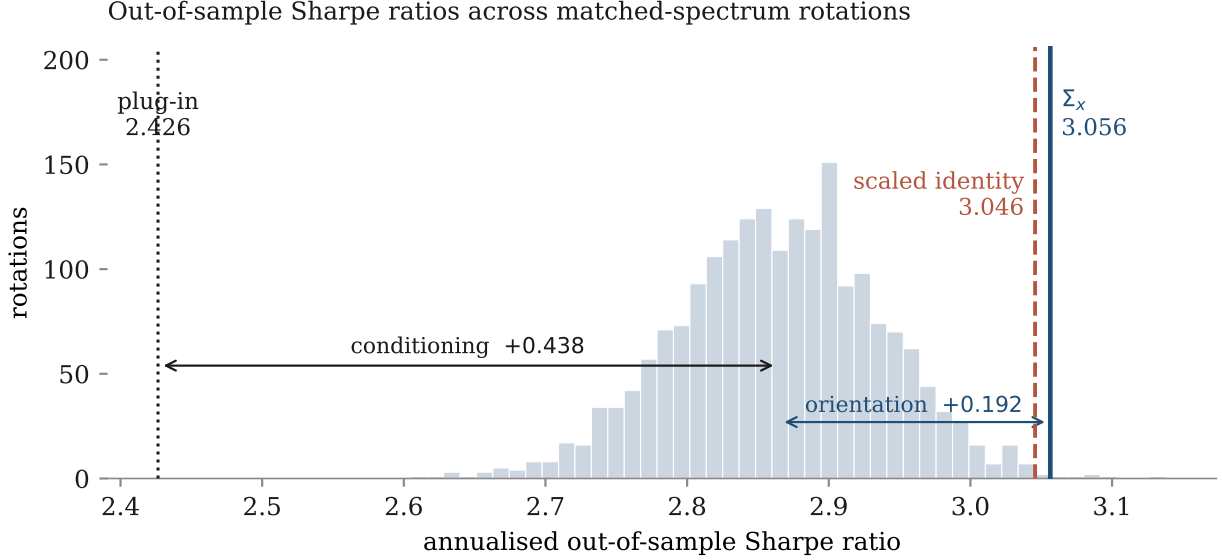


Figure 1: Matched-spectrum rotations. Out-of-sample Sharpe ratios for 2,000 random orthogonal rotations of Σ_x , which leave its spectrum unchanged. Reference lines mark the plug-in estimate, Σ_x , and a scaled identity carrying no characteristic information. 7 of the 2,000 draws reach $SR(\Sigma_x)$.

- unwhitened non-linear shrinkage fails to return a usable estimate on 12.5% of refits, where its trace ratio falls outside $[0.5, 2.0]$, so it is scored on the 503 months it does cover. Every other estimator uses all 575.

The two components. Figure 1 shows the resulting distribution. The mean Sharpe of the 2000 draws is the reference point for both differences, so the conditioning component is $SR(\text{rotation}) - SR(\text{plug-in})$ while the orientation component is $SR(\Sigma_x) - SR(\text{rotation})$, and their spread is the null distribution for the orientation comparison.

Shrinkage intensity. With the shrinkage estimator defined as $\hat{\Sigma} = (1 - w)S + wT$, a further caveat is that the intensity of shrinkage w is set by the analytical rule $w^* = b^2/d^2$, where

$$d^2 = \frac{\|S - T\|_F^2}{p}, \quad b^2 = \frac{\frac{1}{n^2} \sum_t \|x_t x_t' - S\|_F^2}{p}, \quad x_t = f_t - \bar{f}$$

and depends on the target; across refits the median $w_{\Sigma_x}^* = 0.0687$ while $w_{Rotations}^* = 0.0372$. Here n is the length of the training window, x_t the demeaned managed return, and b^2 the Ledoit-Wolf estimate of how noisy the sample is [Ledoit and Wolf, 2004].

Thus, w^* represents a bias-variance tradeoff ratio and b^2 is constant across targets which means variation comes from d^2 . So $\Delta(SR)$ of 0.192 doesn't simply reflect the presence of better eigenvectors, but also that the shrinkage estimator applies greater intensity of shrinkage towards $T = \Sigma_x$ since it is closer to S in the F-norm. The direction of this confound is not neutral, since each of

Table 1: Out-of-sample Sharpe ratios and paired tests against Σ_x .

Target	Class	SR	p vs Σ_x	Bonf.
Σ_x	characteristic	3.056	—	
Scaled identity [Ledoit and Wolf, 2004]	naive	3.046	0.932	
B_{mech}	characteristic	3.041	0.169	
Whitened nonlinear [Ledoit and Wolf, 2020]	spectral	3.038	0.634	
Whitened RIE [Bun et al., 2016]	spectral	2.905	0.000768	✓
Single index [Ledoit and Wolf, 2003]	factor	2.873	0.171	
UPSA [Kelly et al., 2026]	shrinkage	2.849	0.0819	
RIE [Bun et al., 2016]	spectral	2.829	2.51e-05	✓
MAXSER [Ao et al., 2019]	shrinkage	2.795	0.255	
Rotation (single draw)	control	2.763	0.015	
β^* (annual refit)	characteristic	2.654	0.0622	
KNS [Kozak et al., 2020]	shrinkage	2.577	0.0666	
POET (tuned) [Fan et al., 2013]	factor	2.561	3.58e-07	✓
β^* (monthly refit)	characteristic	2.556	0.0669	
Bodnar–Okhrin–Parolya [Bodnar et al., 2023]	shrinkage	2.459	$< 10^{-9}$	✓
Plug-in	none	2.426	$< 10^{-9}$	✓
Kan–Zhou (3-fund) [Kan and Zhou, 2007]	shrinkage	2.425	$< 10^{-9}$	✓
RP-PCA [Lettau and Pelger, 2020]	factor	2.392	9.56e-06	✓
Diagonal volatility	naive	1.855	2.41e-05	✓
μ only	naive	1.165	$< 10^{-9}$	✓
POET [Fan et al., 2013]	factor	0.833	$< 10^{-9}$	✓
Nonlinear shrinkage ^a [Ledoit and Wolf, 2020]	spectral	0.806	1.18e-05	✓
1/ N [DeMiguel et al., 2009]	naive	0.615	$< 10^{-9}$	✓

Paired tests of the Sharpe difference against Σ_x . *Bonf.* marks $p < \alpha = 0.05/22 = 0.00227$. ^a503-month coverage; every other row uses 575. Paired tests use the overlapping months. 12 of 22 comparisons survive.

the four targets in the decomposition improves on the plug-in. So applying more shrinkage towards one of them works in Σ_x ’s favour. I therefore account for the 0.192 as an upper bound on what the eigenvectors contribute by themselves, and call the decomposition operational since it is not a clean experiment on eigenvectors alone.

5 Results

Relative performance. As stated in the introduction, the proposed covariance shrinkage estimator using Σ_x as a target ranks first among the 23 methods tested for Sharpe Ratio performance of the resulting portfolio, with 12 of the 22 Sharpe comparisons surviving Bonferroni corrections at $\alpha = 0.05/22 = 0.00227$. It beats popular benchmarks like whitened RIE, RIE and the tuned POET for this particular application and scope. I should specify that the untuned POET and RP-PCA both select a single factor under their own rules on this panel, so beating them is weaker evidence than beating the tuned POET, which selects 21. Table 1 reports all 23 methods and benchmarks included in the comparison.

While my methodology ranks first, MAXSER, UPSA, KNS ridge, single-index, whitened nonlinear shrinkage and the scaled identity are well established, more general methods that my protocol

fails to beat with statistical certainty.

The +0.438 gain of the rotated null over the plug-in estimator, and the subsequent +0.192 operational orientation component, is a separation of the channels that contribute to the Sharpe gains. $\text{SR}(\Sigma_x)$ is 2.65 SDs above the mean of the null distribution created by the 2000 rotated targets, with 7 of 2000 rotations at or above it, giving a p-value of 0.004 on the $(r + 1)/(n + 1)$ convention. It is worth specifying that this is significant unadjusted but sits above the $\alpha = 0.00227$ I apply to the ladder, and that the orientation test is not in that multiplicity family. There isn't an equivalent null distribution to test the conditioning so the SR gain from conditioning is a point estimate without confidence bounds or a p-value.

Absolute performance. The third target, B_{mech} , is the mechanical benchmark: the training-window average of $c_t X'_t X_t / N_t(N_t - 1)$, where c_t is the cross sectional mean of r_t^2 in month t . It is the object that $\kappa \Sigma_x$ stands in for, and it is built from the same characteristics. Since c_t is the cross sectional second moment of returns and not of residuals, it overstates σ^2 by the factor component, so B_{mech} is an upper bound on the $\kappa \Sigma_x$ term rather than an exact estimate of it. The top three rows of Table 1 show that the three targets land within 0.0155 of each other, which is 0.21 of the null standard deviation, despite having different mechanisms of operation. The scaled identity target has a condition number of 1 by construction, and has no characteristic specific directions. While a ranking is not supported statistically, this reinforces conditioning as the mechanism, with orientation adding a less pronounced effect.

The orientation gain is measured against rotations that keep Σ_x 's own spectrum, so its condition number of 1,714 is held fixed throughout that comparison. The scaled identity instead achieves the same Sharpe with a condition number of 1. So a target with no characteristic specific directions at all is statistically indistinguishable from Σ_x . The rotation null therefore establishes something narrower than it first appears. It shows that Σ_x 's own eigenvectors beat random eigenvectors carrying its own spectrum, and I concede that this is not the same as showing that the performance comes from the characteristic orientation.

The simplicity of a scaled identity achieving similar results points in a negative direction of whether a more structured target is justified within this application. I concede that the evidence points towards generic regularization being sufficiently well suited for its purpose. However, the decomposition of the mechanism and the measurement of the tradeoffs associated are informative.

6 Conclusion

The question this paper tries to answer is what a covariance shrinkage target should be built from for a particular asset space. Since characteristic-managed portfolios are themselves constructed from characteristics, their covariance inherits characteristic structure mechanically. Under the benchmark assumptions of section 3 that inheritance takes an exact form where Σ_f is a quadratic in Σ_x plus a multiple of Σ_x . The justification for shrinking towards Σ_x is therefore a structural

one that a well estimated but economically unrelated matrix would not have, and in the data my proposed methodology ranks first among the methods tested.

The rotation experiment then addresses how much of that performance actually requires the characteristics. Holding the spectrum of Σ_x fixed while randomizing its eigenvectors leaves most of the gain unchanged, with the characteristic directions adding a smaller increment. A scaled identity prior carrying no characteristic information at all finishes close to both of them. My reading of this is that the specific target matters considerably less than I expected when I started.

There are limits which I should state. A5 is the weakest step in the identity and it is not a condition I claim holds exactly. The stability advantage I report for Σ_x is a measurement within the sample period and is not a claim about drift across periods. It is also important to specify that the identity is cleanest in this one asset space, which is why I chose it, and a result about managed portfolios does not establish anything about general equity covariance.

The natural next test is the stock level analogue, $\Sigma_r = X_t M X_t' + \sigma^2 I$, which involves the same M . A second question is how much of this survives a larger set of characteristics, since A1 is a claim about the 132 I use and not about characteristics in general.

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